Midterm 1 Reattempt

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Course: ECE 5200

Problems to regrade:

- 1(a)
- 1(b)
- 2(a–e)
- 3(a–c)

1. a) time reversal and complex conjugation property of DTFT

$$\bar{X}(e^{jw}) = X^*(e^{jw})$$

convolution: $y[n] = x[n] * \bar{x}[n]$

$$Y(e^{jw}) = X(e^{jw})\bar{X}(e^{jw})$$
$$= X(e^{jw}) X^*(e^{jw})$$

b) linearity: $x_1(t) \to y_1(t)$ $\quad x_2(t) \to y_2(t)$

$$y_1(t) + \frac{1}{2}\frac{dy_1(t)}{dt} = x_1(t)$$
$$y_2(t) + \frac{1}{2}\frac{dy_2(t)}{dt} = x_2(t)$$

For constant a, b

$$x(t) = a x_1(t) + b x_2(t)$$
$$y(t) = a y_1(t) + b y_2(t)$$

$$y(t) + \frac{1}{2}\frac{dy(t)}{dt} = a y_1(t) + b y_2(t) + \frac{1}{2}\left(a\frac{dy_1}{dt} + b\frac{dy_2}{dt}\right)$$
$$= a\left(y_1 + \frac{1}{2}\frac{dy_1}{dt}\right) + b\left(y_2 + \frac{1}{2}\frac{dy_2}{dt}\right)$$
$$= a x_1(t) + b x_2(t)$$

$\Rightarrow$ linear

Time Invarince:

$x(t-t_0)$ $\quad y(t-t_0)$

$$y(t-t_0) + \frac{1}{2}\frac{d}{dt}y(t-t_0)$$
$$\frac{d}{dt}y(t-t_0) = y'(t-t_0)$$
$$y(t-t_0) + \frac{1}{2}y'(t-t_0) = x(t-t_0)$$

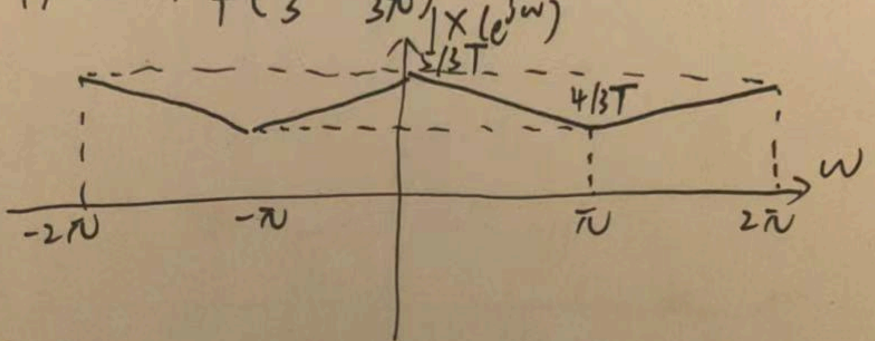
$\Rightarrow$ time-invariant

The system is linear and time-invariant
LTI

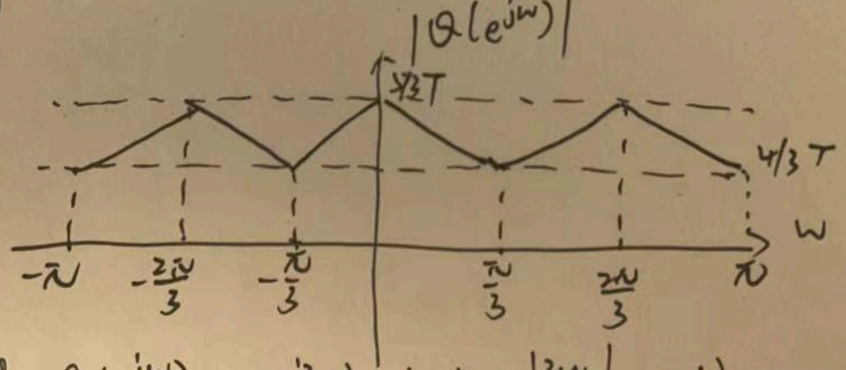
2a) $X(e^{j\omega}) = \frac{1}{T}\left(X_c(j\frac{\omega}{T}) + X_c(j\frac{\omega-2\pi}{T}) + X_c(j\frac{\omega+2\pi}{T})\right)$

$X_c(j\Omega) = \begin{cases} 1 - \frac{|\Omega|}{3\pi/T} & |\Omega| \le 3\pi/T \\ 0 & \text{otherwise} \end{cases}$

$|X(e^{j\omega})| = \frac{1}{T}\left(\frac{5}{3} - \frac{|\omega|}{3\pi}\right), \ -\pi \le \omega \le \pi$



b)



$Q(e^{j\omega}) = X(e^{j3\omega}) = \frac{1}{T}\left(\frac{5}{3} - \frac{|3\omega|_{\text{wrapped}}}{3\pi}\right)$

heigt 5/3T and 4/3T

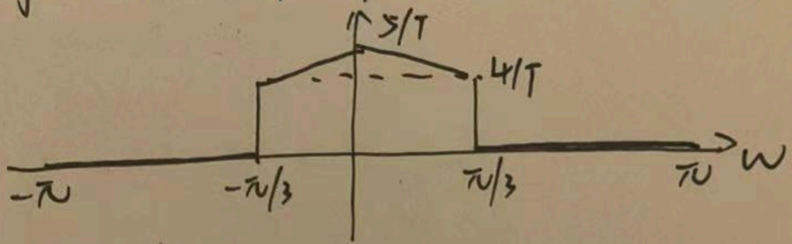
c) $R(e^{jw}) = H(e^{jw})Q(e^{jw})$

$$R(e^{jw}) = \begin{cases} 3Q(e^{jw}) & |w| \le \pi/3 \\ 0, & \pi/3 < |w| \le \pi \end{cases}$$

$$Q(e^{jw}) = X(e^{j3w}) = \frac{1}{T}\left(\frac{5}{3} - \frac{|w|}{\pi}\right)$$

$$R(e^{jw}) = \begin{cases} \frac{1}{T}\left(5 - \frac{3|w|}{\pi}\right) & , |w| \le \pi/3 \\ 0 & , \pi/3 < |w| \le \pi \end{cases}$$

height: $5/T$, $4/T$ $|R(e^{jw})|$



d) $v[n] = r[n-1]$

$V(e^{jw}) = e^{-jw} R(e^{jw})$

$$V(e^{jw}) = \begin{cases} e^{-jw}\frac{1}{T}\left(5 - \frac{3|w|}{\pi}\right) & |w| \le \pi/3 \\ 0 & \pi/3 < |w| \le \pi \end{cases}$$

e) $Y(e^{jw}) = \frac{1}{3}\sum_{k=0}^{2} V(e^{j(w-2\pi k)/3})$

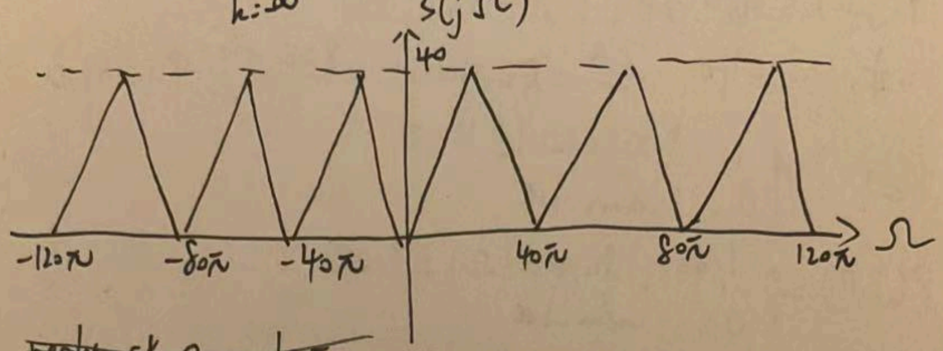
$$Y(e^{jw}) = \frac{1}{3}V(e^{jw/3})$$
$$= \frac{1}{3}e^{-jw/3}R(e^{jw/3})$$
$$= \frac{1}{3T}e^{-jw/3}\left(5 - \frac{|w|}{\pi}\right)$$

$X(e^{jw}) = \frac{1}{3T}\left(5 - \frac{|w|}{\pi}\right)$

$Y(e^{jw}) = e^{-jw/3}X(e^{jw})$

$y[n] = x_c\left(nT - \frac{T}{3}\right)$

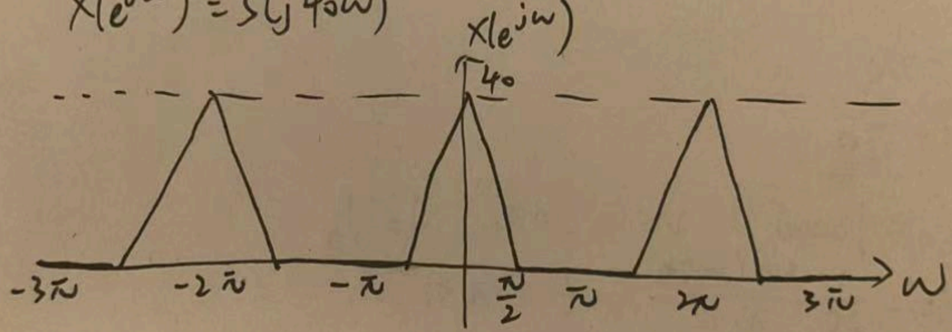
3 a) $S(j\Omega) = 40 \sum_{k=\infty}^{\infty} X_c(j(\Omega - 80\pi k))$



~~peak at $\Omega = \pm 40\pi$~~,

b) $T = 1/40$

$X(e^{j\omega}) = S(j\,40\omega)$



c) $y(t) = s(t) * p(t)$

$Y(j\Omega) = S(j\Omega)P(j\Omega) = X_c(j\Omega)$

$\frac{80\pi + 120\pi}{2} = 100\pi$

$T = \frac{1}{40}$ $\frac{1}{T} = 40$ $\frac{2\pi}{T} = 80\pi$

$P(j\Omega) = \begin{cases} T, & 80\pi \le |\Omega| \le 120\pi \\ 0, & \text{otherwise} \end{cases}$

$P(j\Omega) = \begin{cases} 1/40, & 80\pi \le |\Omega| \le 120\pi \\ 0, & \text{otherwise} \end{cases}$

$\Omega_0 = 100\pi$ (center)

$W = (120 - 100)\pi = 20\pi$

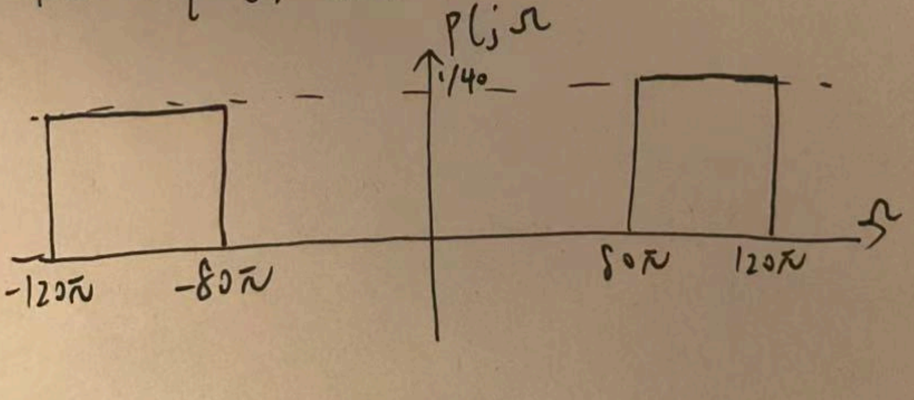
$\frac{A}{2} = \frac{1}{40}$

$A = \frac{1}{20}$

$\Omega_0 = 100\pi$ $W = 20\pi$ $A = \frac{1}{20}$

$p(t) = \frac{1}{20} \frac{\sin(20\pi t)}{\pi t} \cos(100\pi t)$

$P(j\Omega) = \begin{cases} 1/40, & 80\pi \le |\Omega| \le 120\pi \\ 0, & \text{otherwise} \end{cases}$



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Spring 2026

The Ohio State University
ECE 5200

Midterm #1

Friday, February 20, 2026

Name: Yangziqian Guo

ID: 500673783

Score: 41.5

Problem	Pts.	Score
1	40	20
2	50	21.5
3	30	0
Total	120	41.5

I certify that this is my sole work and I haven't discussed any problem with other person.

Signature: Yangziqian Guo

1. Please print your name and student ID number.
2. Sign on the above statement.
3. For full credit, you should show all of your work and explain your reasoning.

GOOD LUCK!

Problem 1 (40 Pts) **(a)** (20 Pts) Let $x[n]$ be a discrete-time signal, and define

~~10~~ 10

$$\tilde{x}[n] = x^*[-n],$$

where $x^*[n]$ denotes the complex conjugate of $x[n]$. Let

$$y[n] = x[n] * \tilde{x}[n].$$

Express the DTFT $Y(e^{j\omega})$ of $y[n]$ in terms of the DTFT $X(e^{j\omega})$ of $x[n]$.

$y[n] = x[n] * \tilde{x}[n]$

$Y(e^{j\omega}) = X(e^{j\omega}) X^*(\cancel{e}\,e^{jw\cancel{0}})$

which property ?

no - sign

wrong answer   no derivation

(b) (20 Pts) Consider the continuous-time system defined by

10

$$y(t) + \frac{1}{2} \cdot \frac{dy(t)}{dt} = x(t)$$

where $x(t)$ is the input and $y(t)$ is the output. Determine whether this system is linear, time-invariant, both, or neither. Provide a clear justification for each property you assess.

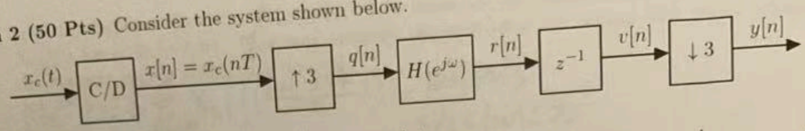
not linear $\quad 2y(t) + \frac{dy(t)}{dt} \neq y(2t) + \frac{1}{2}\frac{dy(2t)}{dt}$

$2x(t) \neq x(2t)$

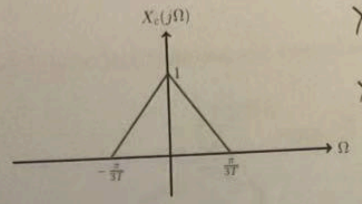
not time-invariant $\quad y(t-t_0) = x(t-t_0) - \frac{1}{2}\frac{dy(t-t_0)}{dt}$

It is LTI   try freq. domain approach

Problem 2 (50 Pts) Consider the system shown below.



Suppose that $X_c(j\Omega)$ is given as in the following plot.



$X_c(j\Omega) = \left|\frac{3T}{\pi}\Omega\right| - 1$

$X(e^{j\omega}) = \frac{1}{T}\sum_{-\infty}^{\infty} X_c\left(j\left|\frac{3T}{\pi}\left(\frac{\omega-2\pi k}{T}\right)\right| + \right.$

$(j(3\omega - 6k| - 1))$

$X_c(0) = 1$

The frequency response of the filter satisfies

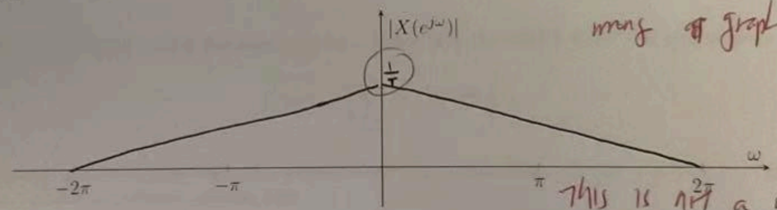
$$H(e^{j\omega}) = \begin{cases} 3, & |\omega| \le \pi/3, \\ 0, & \pi/3 < |\omega| \le \pi. \end{cases}$$

(a) (10 Pts) Recall that $X(e^{j\omega})$ can be expressed in terms of $X_c(j\Omega)$ as

5

$$X(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - 2\pi k}{T}\right)\right).$$

Plot the magnitude $|X(e^{j\omega})|$ over the interval $-2\pi \le \omega \le 2\pi$. Label both horizontal and vertical axes.



wrong graph

this is not a valid DTFT

not 2π-periodic

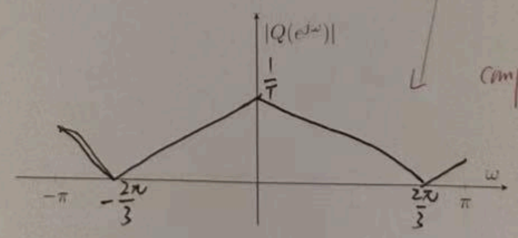
(b) (10 Pts) Recall that the DTFT $Q(e^{j\omega})$ of the upsampled signal $q[n]$ is given by

9

$$Q(e^{j\omega}) = X(e^{j\omega 3}).$$

Plot the magnitude $|Q(e^{j\omega})|$ over the interval $-\pi \le \omega \le \pi$. Label both horizontal and vertical axes.

compression



still not a valid DTFT

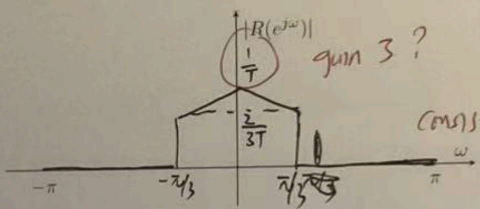
only $k=0$ term survives

(c) (10 Pts) Express $R(e^{j\omega}) = H(e^{j\omega})Q(e^{j\omega})$ in terms of the continous-time Fourier transform $X_c(j\Omega)$ for $\omega \in [-\pi, \pi)$. You may simplify the expression by noting how many triangular components appear within this interval.

7.5

$$R(e^{j\omega}) = \begin{cases} \frac{3}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{3\omega-2\pi k}{T}\right)\right), & |\omega| \le \pi/3 \\ 0 & \pi/3 < |\omega| \le \pi \end{cases}$$

Plot the magnitude $|R(e^{j\omega})|$. Label both horizontal and vertical axes.



(d) (10 Pts) Note that $v[n]$ is a one-sample delayed version of $r[n]$. First, express the DTFT $V(e^{j\omega})$ of $v[n]$ in terms of the DTFT $R(e^{j\omega})$ of $r[n]$. Then, by using the result from Part (c), rewrite $V(e^{j\omega})$ in terms of $X_c(j\Omega)$ for $\omega \in [-\pi, \pi)$.

6

z^{-1}

(e) (10 Pts) Recall that after downsampling by a factor of 3, the DTFT $Y(e^{j\omega})$ of $y[n]$ is given by

0

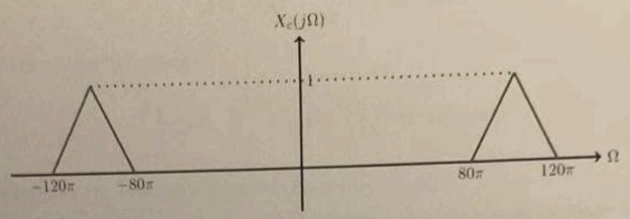
$$Y(e^{j\omega}) = \frac{1}{3}\sum_{k=0}^{2} V\left(e^{j\frac{\omega-2\pi k}{3}}\right).$$

By using the result from Part (d), express $Y(e^{j\omega})$ in terms of $X_c(j\Omega)$ for $\omega \in [-\pi, \pi)$. Then, based on this expression, explain why

$$y[n] = x_c\left(nT - \frac{T}{3}\right).$$

In other words, show that $y[n]$ corresponds to a fractional-delay version of the original sampled signal $x[n]$, delayed by one-third of a sample interval.

Problem 3 (30 Pts) Consider a continuous-time signal $x_c(t)$ whose Fourier transform $X_c(j\Omega)$ is given as in the following plot.



(a) (10 Pts) Let

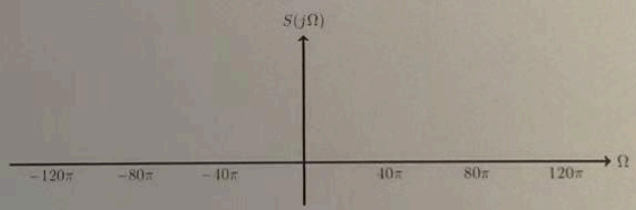
$$s(t) = x_c(t) \sum_{k=-\infty}^{\infty} \delta(t - nT), \quad \text{where} \quad T = \frac{1}{40}.$$

0

Recall that the CTFT $S(j\Omega)$ of $s(t)$ can be expressed in terms of $X_c(j\Omega)$ as

$$S(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\Omega - \frac{2\pi k}{T}\right)\right).$$

Plot $S(j\Omega)$. Label both horizontal and vertical axes.



(b) (10 Pts) Let

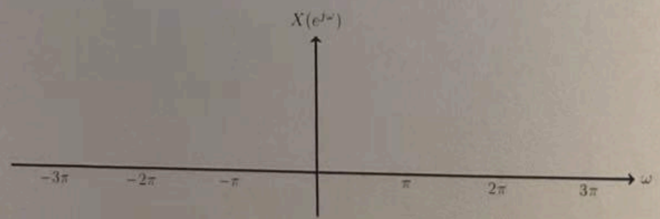
$$x[n] = x_c(nT), \quad \text{where} \quad T = \frac{1}{40}.$$

0

Recall that the DTFT $X(e^{j\omega})$ of $x[n]$ can be expressed as

$$X(e^{j\omega}) = S\left(j\frac{\omega}{T}\right).$$

Plot $X(e^{j\omega})$. Label both horizontal and vertical axes.



(c) (10 Pts) Design the interpolation kernel

$$p(t) = A \cdot \frac{\sin(Wt)}{\pi t} \cdot \cos(\Omega_0 t)$$

so that the reconstructed signal

$$y(t) = \sum_{k=-\infty}^{\infty} x[n]p(t-kT) = s(t) * p(t)$$

exactly reproduces the original continuous-time signal $x_c(t)$.

i) Determine the parameters A, W, and Ω_0 required to ensure exact reconstruction.

ii) Provide an analytic expression for $p(t)$ using these parameter values.

iii) Plot the Fourier transform $P(j\Omega)$ of the interpolation kernel. Label both the horizontal and vertical axes.